Sebastian F. Otarola-Bustos, PhD

Rockville, MD · sfotarol@gmail.com · linkedin.com/in/sotarolab

My background spans water resources engineering, atmospheric fluid dynamics, and applied machine learning, in industry and in research.

Experience

Principal Scientist

Aug 2022 – Present

Hazen, Silver Spring, MD

- Develop and calibrate hydraulic and hydrologic models for utilities and water districts across the US, supporting regulatory compliance, flood analysis, water supply management, and capital improvement planning in urban and natural systems.
- Lead machine learning work in classification, regression, and anomaly detection: wet-weather influent flow forecasting, urban flooding detection, and water demand forecasting, following version-controlled development practice and modern ML frameworks.
- Contributed to a framework for building and calibrating parsimonious (sparse-parameter) hydrological models in Python, aimed at flooding in stormwater systems.
- Led data analysis and web tool development for a research-foundation study of water quality in quarry lakes, screening about 500,000 US lakes from in-situ and remote sensing records as alternative supply storage.

Industry

Research & Independent Projects

2025 – Present

Self-directed

- Collaborative research with Chile and US researchers training a convolutional neural network to downscale CMIP6-based projections of extreme precipitation and temperature over Chile, and investigating the physical mechanisms driving the changes so the method can transfer to other regions.
- Applying Long Short-Term Memory (LSTM) and diffusion algorithms to predict flow discharges in US west coast basins, stress-tested on atmospheric-river flood events and benchmarked against process-based models (GR4J and CemaNeige) across hydrological regimes.
- Maintain two public platforms with live ingestion from operational forecast models and observation networks, plus a bias-correction pipeline that calibrates the forecasts: a US heat-wave tracker covering major airports, and an atmospheric river tracker following weather stations and flood alerts in Chile.

Independent

PhD Researcher, Atmospheric Fluid Dynamics

Aug 2016 – May 2022

University of Notre Dame

- Participated in two major field experiments: the Wind Forecast Improvement Project 2 (WFIP2) in the Columbia River Gorge (OR) and the Sundowner Winds Experiment (SWEX) in Santa Barbara (CA), deploying flux towers and Doppler lidars and processing their observations to study the skill of weather forecasting models in complex terrain.
- Published the WFIP2 results in *Boundary-Layer Meteorology*, on the subgrid variability of surface layer turbulence in complex terrain from 18 months of observations, benchmarked against the WRF model.
- Collaborated on peer-reviewed publications in boundary layer meteorology and wind energy, produced open datasets published in the DOE Atmosphere to Electrons archive, and presented at national and international conferences.

Academic

Education

PhD, Civil & Environmental Engineering & Earth Sciences

2016 – 2022

University of Notre Dame

Atmospheric fluid dynamics: turbulence and complex-terrain boundary-layer flows. Advised by Dr. Harindra Joseph Fernando.

EFM Lab profile →

B.S., Water Resources Engineering

2009 – 2015

Pontificia Universidad Católica de Chile

Technical Skills

Programming & Computational Stack
Python Programming • MATLAB • GitHub Workflows • SQL, SQLite, PostgreSQL • Docker • C++ • Linux • HTML • Cloud Computing • LLMS • Agentic AI Workflows • ArcGIS Pro • H&H Modeling Software

Atmospheric & Climate Models
NWP Forecasting Systems (GFS, ECMWF IFS/AIFS, NAM, HRRR, WRF) • CMIP6 Project • ERA 5 • Remote Sensing (Sentinel 1, Sentinel 2, GOES)

Teaching & Mentoring

Engineering mentoring at Hazen and Sawyer since 2022, and graduate teaching assistantships in environmental hydrology and fluid mechanics at Notre Dame. See **Teaching** for the full record.